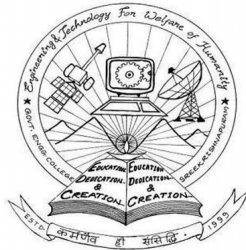


ITT464 COMPUTER GRAPHICS

Module 4: 2-D Clipping & 3-D Geometric Transformations



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Acknowledgements

- Some pictures are taken from the Internet using Google search.
- Some are taken from the book Computer Graphics by Hearn & Baker
- Wikipedia also referred.

Meeting 10

What's in Store?

Syllabus

Module 1: Graphics Systems (8 Hours)
Raster Scan & Random Scan systems. Output Primitives – Line Drawing Algorithms (DDA, Bresenham), Bresenham's circle generation algorithm. Filled Area Primitives – Scan Fill, Flood Fill, Boundary Fill. Inside outside tests.
Module 2: Display Technologies(6 Hours)
Working principle behind CRT, LCD, Plasma, LED, OLED, AMOLED, E-Paper displays.
Module 3: 2-Dimensional Geometric Transformations (8 Hours)
Basic Transformations, Reflection & Shear, Homogenous Matrix representation of transformations, Composite Transformations.
Module 4: 2-D Clipping & 3-Dimensional Geometric Transformations (6 Hours)
2-D Clipping: Point Clipping, Cohen-Sutherland Line Clipping Algorithm, Sutherland-Hodgeman Polygon Clipping Algorithm. 3-Dimensional Geometric Transformations: Basic transformations, Composite 3D transformations (Basics only)
Module 5: Visible Surface Detection Methods & Colour Models (7 Hours)
Visible Surface Detection Methods: Back Face Detection, Depth Buffer, A-Buffer, Scan line, Depth sorting methods. Colour Models: RGB Model, CMY Model, HSV Model

Text Books

1. Donald Hearn, Pauline Baker, Computer Graphics – C Version, Pearson Education, 2nd Edition, 2002
2. Amarendra N' Sinha and Arun D Udai, "Computer Graphics", The McGraw-Hill

We Saw & Will See

Till Now We Saw...

- Module 1 : Graphic Systems
 - Raster Scan & Random Scan
 - Line Drawing Algorithm : DDA, Bresenham's
 - Bresenham's Circle(Midpoint Circle)
 - Filling Algorithms: Scan Line Fill , Flood Fill , Boundary Fill
 - Inside-Outside Tests : Odd-Even, Non-zero Winding
- Module 3: 2-Dimensional Geometric Transformations
 - Basic Transformations : Translation, Rotation and Scaling
 - Homogeneous Matrix Representations
 - Composite Transformations
 - Reflection and Shear

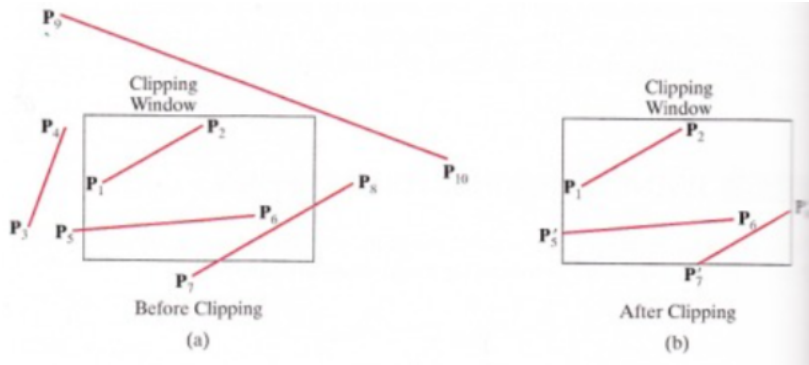
Today We Will See...

- Module 4: 2-D Clipping & 3-D Geometric Transformations
 - Point and Line Clipping

Point Clip

Clip against a window with $x_{w_{min}}$, $x_{w_{max}}$, $y_{w_{min}}$, $y_{w_{max}}$

Line Clipping Algorithm



Line Clipping

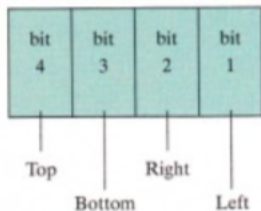
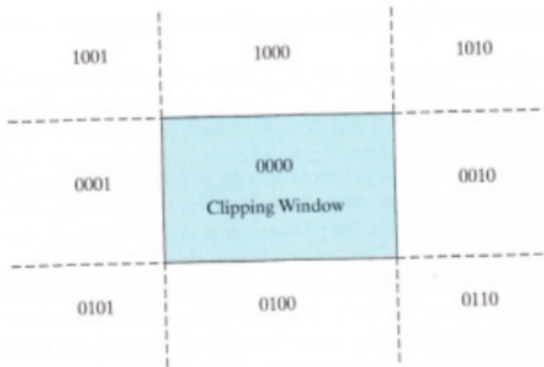


FIGURE 6-12 A possible ordering for the clipping-window boundaries corresponding to the bit positions in the Cohen-Sutherland endpoint region code.

Line Clipping Clipping



Meeting 11

We Saw & Will See

Till Now We Saw...

- Module 1 : Graphic Systems
 - Raster Scan & Random Scan
 - Line Drawing Algorithm : DDA, Bresenham's
 - Bresenham's Circle(Midpoint Circle)
 - Filling Algorithms: Scan Line Fill , Flood Fill , Boundary Fill
 - Inside-Outside Tests : Odd-Even, Non-zero Winding
- Module 3: 2-Dimensional Geometric Transformations
 - Basic Transformations : Translation, Rotation and Scaling
 - Homogeneous Matrix Representations
 - Composite Transformations
 - Reflection and Shear
- Module 4: 2-D Clipping & 3-D Geometric Transformations
 - Point, Line Clipping

Today We Will See...

- Module 4: 2-D Clipping & 3-D Geometric Transformations
 - Polygon Clipping

Clipping Algorithms-Polygon

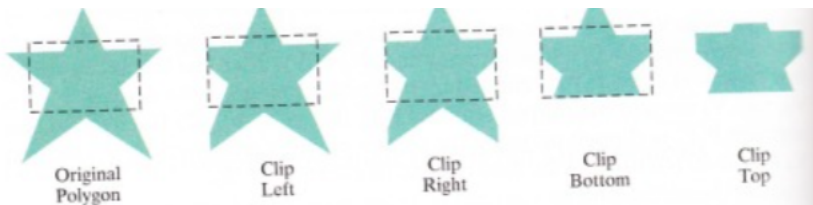


FIGURE 6-23
boundaries.

Processing a polygon fill area against successive clipping-window

Clipping Algorithms-Polygon

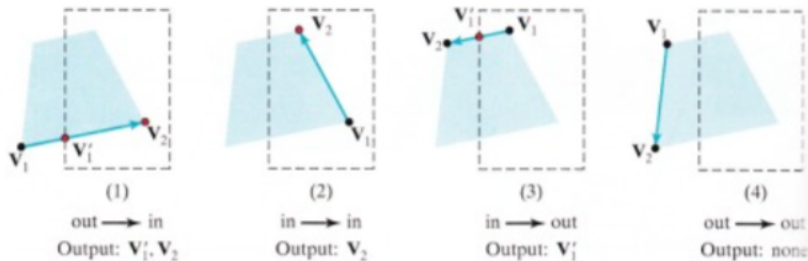


FIGURE 6-26 The four possible outputs generated by the left clipper, depending on the position of a pair of endpoints relative to the left boundary of the clipping window.

Clipping Algorithms-Polygon

